Fiscal Year:	FY 2013	**Task Last Updated:**	FY 11/05/2012
PI Name:	Nelson, Gregory A. Ph.D.		
Project Title:	NSCOR: Charged Particle Radiation and Resultant Oxidative Stress Elicit Deleterious Functional Changes in the Central Nervous System		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	grnelson@llu.edu	**Fax:**	FY 909-558-4035
PI Organization Type:	UNIVERSITY	**Phone:**	909-558-8364
Organization Name:	Loma Linda University		
PI Address 1:	Basic Sciences, Div. Radiation Research 11175 Campus Street		
PI Address 2:	Chan Shun Pavilion, Room A-1024		
PI Web Page:			
City:	Loma Linda	**State:**	CA
Zip Code:	92350-1700	**Congressional District:**	31
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 NSCOR Space Radiation NNJ08ZSA003N
Start Date:	01/01/2010	**End Date:**	12/31/2014
No. of Post Docs:	8	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	November 2012 report: N Komarova and D Wodarz's tasks were not renewed for contract years 4 & 5.		
COI Name (Institution):	Soltesz, Ivan (University of California, Irvine) Wodarz, Dominik (University of California, Irvine) Limoli, Charles (University of California, Irvine) Vlkolinsky, Roman (Loma Linda University) Obenaus, Andre (Loma Linda University) Fike, John (University of California, San Francisco) Komarova, Natalia (University of California, Irvine) Spigelman, Igor (UCLA School of Dentistry) Raber, Jacob (Oregon Health & Science University)		
Grant/Contract No.:	NNX10AD59G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The proposed project will investigate responses of the mouse brain to several radiation exposure regimes corresponding to galactic cosmic rays and solar particle events. The project will characterize behavioral, electrophysiological, biochemical and structural changes both in vivo and in vitro. Radiation-elicited oxidative stress mechanisms will be investigated as important mediators of the biological responses in both wild type and mutant animals. In addition to characterization of normal tissue responses, the potential enhancement of three late risks by radiation exposure will be explored. These are seizures, ischemic injury and traumatic injury. Mathematical models of the neurotoxicity responses will developed for use in estimating risks and fitted to project-specific data and previously-published observations. The aims of the proposal will address HRP risk 1.2 and at least five of the specific gap questions identified by NASA: CNS-1, -3, -5, -6 and -11.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	Understanding the responses of the central nervous system to charged particle radiation has two main benefits – risk assessment and optimization of radiotherapy. The general population and aviation crews are exposed to charged particles during cross country and trans-oceanic air travel at high altitudes as well as to radon in their homes and workplaces. While these exposures are not as high as those of astronauts they are not insignificant. Cancer risks have traditionally been the focus of health risk estimates but the new awareness of sensitive targets in the brain such as neurogenic centers in the hippocampus raises question about the negative effects of such radiation on late neurodegenerative diseases and potential decline of cognitive abilities. This project directly addresses the cognitive and neurogenesis issues and indirectly addresses late neurodegenerative effects which share reactive oxygen chemistry with radiation as underlying mechanisms. Alzheimer's, Huntington's, and in particular, ALS diseases, are of interest. In the last ten years, the use of charged particles in radiation therapy has grown dramatically with proton and carbon ion treatment centers being established in the U.S., Europe, and Asia. Always limiting the effectiveness of radiotherapy is the response of normal tissue near the tumor site. The precise dose localization associated with charged particle therapy makes it ideal for tumors of the head and neck where careful avoidance of critical structures such as the optic nerve is critical. Knowledge of the dose "tolerance" of brain tissue and the reactions to relatively low doses is critical in optimizing these modalities of treatment.		

	The CNS NSCOR focuses on the functional consequences of exposure of the central nervous system to charged particles. This work builds on previous results by our team members that demonstrated charged-particle radiation induced alterations in synaptic function, neurogenesis, behavior, microvasculature organization, and structural modification of the hippocampus. The approaches used explicitly examine the roles of reactive oxygen species and neurogenesis in mediating the effects of radiation exposure. They also examine the functional properties of neurons and local circuits and will link these properties to the fidelity of behavioral output. Computational models of the hippocampus in three dimensions, informed by experimental measurements, will enable us to manipulate properties of neurons and circuits to provide insight into network behavior. Finally, aspects of radiation quality and dose rate relevant to spaceflight exposures will be assessed for the endpoints above. Radiation exposure protocols include single, acute whole body exposures to protons, silicon, and iron ions. To date 631 animals have undergone radiation exposure and analysis. Complex exposure regimens include: 1) adaptation to priming doses of protons followed by challenge doses of iron ions and 2) dose rate will be assessed using fractionated iron ions. The animal models are 10 week old C57Bl/6 males and transgenic MCAT males which over-express human catalase in mitochondria. To date we have irradiated mice with protons, silicon, and iron ions as well as protons and iron ions in combination. Animals are evaluated at 90 days post irradiation. In vitro models are cultured murine and human neural stem cells irradiated with multiple ions at different energies and are evaluated at times from days to months. Highlights from studies during 2011 and 2012 Human neural stem cells were irradiated with several ions at energies from 150-1000 MeV/n in the dose range 0.75 to 30 cGy. Data suggest that significant changes in oxidative stress occur that alter radiosensitivity, proliferation and differentiation. Overall trends indicate that changes in oxidative stress (observed days to weeks post-irradiation) correlate with particle LET with iron ions eliciting the largest magnitude and most persistent changes in stress biomarkers including antioxidant enzyme expression levels. Baseline characterization of MCAT mice vs. wild type mice was conducted prior to irradiation measurements. Prior measurements of anxiety levels in 5-6 month wild type and MCAT animals using elevated zero maze suggest that MCAT animals exhibited reduced measures of anxiety. The hippocampus-dependent contextual fear conditioning paradigm, supplemented by novel object recognition and water maze paradigm were used to assess cognition. These experiments showed no genotypic differences for novel object recognition. For maze performance MCAT animals showed a trend for better performance relative to wild type suggesting lower oxidative stress may be beneficial. With contextual fear conditioning there were no genotypic differences in the freezing response but a greater sensitivity to shock stimulus may be present in MCAT animals. Together these observations provide a baseline for inter-comparison of the two genotypes in preparation for irradiation experiments. Results with irradiation with iron ions suggested that the novel object recognition paradigm is more sensitive than the contextual fear conditioning and avoids anxiety-related issues from negative motivation methods. MCAT mice were found to be significantly impacted by proton doses of 0.1 and 0.5 Gy. Analysis of neurogenesis and Arc expression in these animals indicated that neurogenesis was not inhibited but newly-born activated microglia were significantly elevated at 0.1 Gy with some regional dependence within the dentate gyrus. Analysis of neurogenesis suggested a dose dependent decrease in neurogenesis following iron ion exposure at 1 Gy but not 0.25 Gy. In the adaptive response irradiation regimen, with combinations of protons and iron ions, novel object recognition was impaired with 0.1 Gy protons or 0.1 Gy protons + 0.5 Gy iron but not with 0.5 Gy iron alone. A negative correlation between newly-born activated microglia and novel object recognition was observed for exposures using protons and iron ions suggesting that neuroinflammation contributes to the cognitive injury in a dentate gyrus region specific pattern. There were no reductions in neurogenesis and Arc expression under these conditions. Patch clamp recordings on pyramidal neurons of the CA1 hippocampus field were conducted on mice irradiated with protons, silicon and iron ions. Twelve parameters were measured for CA1 pyramidal cells and two parameters were found to be altered by radiation exposure with protons being the most effective radiation species. Two interesting parameters were the input resistance and the resting membrane potential. The changes observed are consistent with the neurons become less responsive post irradiation. An estimate of the number of synaptic vesicles required to be released for signal transmission was made and a 40% increase was observed for 1 Gy protons. Changes in persistent sodium currents and total membrane currents were also observed after proton exposure and differed between wild type and MCAT animals. Silicon and iron irradiated animals showed only minor alterations in amplitudes of miniature excitatory and inhibitory post synaptic currents.
Task Progress:	Patch clamp recordings on granule neurons of the Dentate Gyrus (DG) hippocampus field were conducted on mice irradiated with protons and heavy iron ions. Previous experiments with protons indicated changes in miniature inhibitory post synaptic currents so Western blots were performed to detect changes in three GABAA receptor subunits. No statistically significant differences in receptor quantities or stoichiometry were detected which may mean that receptor function has been altered instead. Patch clamp measurements on silicon-irradiated animals failed to detect any differences in mEPSCs or mIPSCs at 0.25 and 1 Gy relative to sham. With iron irradiation there was a trend to increase mIPSC amplitudes with dose but no effect was observed on frequency. By contrast, significant increases in AMPA receptor-mediated mEPSC frequency were observed without affecting amplitude. This suggests effects on presynaptic glutamate release mechanisms. Using extracellular field recordings with microelectrode arrays we analyzed functional changes in the CA1 hippocampal synaptic network of C57Bl/6J mice triggered by whole body irradiation with protons, iron and silicon radiation. The most complete set of data is represented by silicon irradiated animals, while iron experiment data are undergoing further analyses and statistical evaluations. The significance of deficits reported in proton irradiated mice are currently being evaluated. The effects observed with protons contrasts with previously described high-LET effects of 56Fe irradiation where dose-dependent increases of fEPSPs were observed at doses 1-4 Gy. In the DG neuronal field the effect of radiation appeared to be opposite, perhaps due to preferential effect of radiation on feed-forward inhibitory neurons that exert control over granular cell excitability. This effect is now well supported by our intracellular findings on reduced inhibitory synaptic currents (mIPSC) in the DG described in previous sections. Clearly, there are regional differences in radiation sensitivity to protons between CA1 and DG neurons. Effects of silicon irradiation tested in CA1 neurons in the ventral hippocampus point toward damage in dendro-somatic functional coupling (E-S coupling) with consequence of altered signaling of the hippocampal neurons to other brain structures. Similar deficits were described in Guinea Pig hippocampal slices acutely exposed to high dose of photon radiation, however it has not been observed in our previous studies with head only irradiations with iron (1 mo post irradiation). Modeling activities with simplified examinations of cell population dynamics for neural precursor cells were begun with the intent to probe the effects of cell communication in modulating adaptive responses. However, these activities were not productive and were terminated. New modeling activities using a high fidelity three-dimensional model of the hippocampus have been recently initiated and will allow simulation of network activities perturbed by alteration of neuron parameters along linear tracks simulating the passage of charged particles. In addition to the publications listed in the Bibliography section, the researchers participated in the following: Fike JR. "Ionizing Irradiation and the CNS." Invited Talk. 58th Annual Meeting of the Radiation Research Society, San Juan, Puerto Rico, September 30 – October 3, 2012. Limoli CL. "This House Believes that Cancer Stem Cells and their Normal Counterparts Dictate the Outcome of Clinical Radiotherapy." Invited Talk. Debate D1. 58th Radiation Research Society Annual Meeting, San Juan, Puerto Rico, September 30 – October 3, 2012. Vlkolinsky R. "CNS Damage. In: Questions in Space Radiation Research." Workshop WS2. 58th Annual Meeting of the Radiation Research Society, San Juan, Puerto Rico, September 30 – October 3, 2012. Fike JR. "Neurogenesis; Radiation Effects on Neurons and Stem Cells." Invited Lectures. Summer School Texbook, NASA Space Radiation Summer School. Brookhaven National Laboratory, Upton, NY. June, 2012. Nelson G. "Space Flight Measurements; Haematopioetic and Immune Responses; Microgravity Effects." Three invited lectures. NASA Space Radiation Summer School. Brookhaven National Laboratory, Upton, NY. June, 2012 Nelson GA. "NSCOR Program Projects" NASA Central Nervous System Risk External Review. League City, Texas. NASA Report. Jan-2012 Limoli C, Nelson G, Raber, J. "Lecture to a review panel on progress in CNS and non-cancer research." Invited panelists. 23nd Annual NASA Space Radiation Investigators' Workshop 2012, Durham, North Carolina, July 2012.

Bibliography Type:	Description: (Last Updated: 03/26/2026)
Abstracts for Journals and Proceedings	Acharya M, Rosi S, Christie L-A, Limoli CL. "Mechanisms underlying stem cell mediated restoration of radiation-induced cognitive decrements." 58th Annual Meeting of the Radiation Research Society, San Juan, Puerto Rico, September 30 – October 3, 2012. 58th Annual Meeting of the Radiation Research Society, San Juan, Puerto Rico, September 30 – October 3, 2012. Presentation number: PS5-38. Abstract accessible at: http://www.abstractsonline.com/Plan/ViewAbstract.aspx?sKey=f8c8e816-7bf5-4085-94f8-f076ca29978a&cKey=11e41e27-8e23-46ab-b868-5a8927953b35&mKey={1514526B-EDA6-42BC-948B-C58023E5A94E} ; accessed 11/15/12. , Oct-2012
Abstracts for Journals and Proceedings	Allen A, Habdank-Kolaczkowski J, Chakraborti A, Sharma S, Baure J, Allen B, Rosi S, Raber J, Fike JR. "Juvenile radiation exposure induces a predisposition to cognitive alterations following mild traumatic brain injury in adulthood." 58th Annual Meeting of the Radiation Research Society, San Juan, Puerto Rico, September 30 – October 3, 2012. 58th Annual Meeting of the Radiation Research Society, San Juan, Puerto Rico, September 30 – October 3, 2012. Presentation number: PS1-28. Abstract accessible at: http://www.abstractsonline.com/Plan/ViewAbstract.aspx?sKey=ad334a4c-47f6-4999-9ddf-153546ba4fc9&cKey=a55397c0-0994-48d3-9073-f38139027f13&mKey={1514526B-EDA6-42BC-948B-C58023E5A94E} ; accessed 11/15/12. , Oct-2012
Abstracts for Journals and Proceedings	Haley GE, Yeiser L, Olsen R, Raber J. "Short-term effects of 56Fe irradiation on cognitive function in C57Bl6/J female and male mice." 41st Annual Society for Neuroscience Meeting, Washington D.C., November 12-16, 2011. 41st Annual Society for Neuroscience Meeting, November, 2011. Program#/Poster#: 410.03/ZZ64. Abstract available at http://www.abstractsonline.com/Plan/ViewAbstract.aspx?sKey=dcda5d82-87b9-473f-b9cd-98ff4b69b827&cKey=792677bd-dfd3-46b6-b68a-09bf3b6db193&mKey={8334BE29-8911-4991-8C31-32B32DD5E6C8} ; accessed 11/15/2012. , Nov-2011
Abstracts for Journals and Proceedings	Limoli C, Kumar V, Tseng B, Tran K, Craver B, Giedzinski E. "Redox alterations and changes to neuronal anatomy caused by low dose irradiation." 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. Invited Talk 8034. 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. , Jul-2012
Abstracts for Journals and Proceedings	Marty VN, Minassian N, Cohen T, Nelson G, Spigelman I. "Radiation-induced alterations in synaptic transmission of dentate granule neurons depend on the dose and species of charged particles." 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. Poster 8100. 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. , Jul-2012
Abstracts for Journals and Proceedings	Nelson GA, Fike JR, Limoli C, Obenaus A, Raber J, Spigelman I, Vikolinsky R. "CNS NSCOR Overview." 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. Invited Talk #8061. 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. , Jul-2012
Abstracts for Journals and Proceedings	Raber J, Belknap J, Kleiman NJ, Hall EJ, Ray FA, Fallgren CM, Weil MM. "Characterization of the Tumor Spectrum Arising in HZE Ion Irradiated Outbred Mice." 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. , Jul-2012
Abstracts for Journals and Proceedings	Raber J, Haley GE, Johnson L, Yeiser L. "Early Effects of Alpha-Lipoic Acid on Cognition of Sham-Irradiated and 56Fe-Irradiated C57BL/6J Female and Male Mice." 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. , Jul-2012
Abstracts for Journals and Proceedings	Raber J, Allen A, Baure J, Habdank-Kolaczkowski J, Davis MJ, Olsen R, Yeiser L, Rosi S, Fike JR. "Charged particle radiation, reactive oxygen species, and CNS function." 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. Poster presentation 8057. 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. , Jul-2012
Abstracts for Journals and Proceedings	Schneider C, Case M, Soltesz I. "Enlargement and improvement of a computational model of the rat dentate gyrus." 19th Joint Symposium on Neural Computation, Riverside, CA, June 2, 2012. Poster presentation. 19th Joint Symposium on Neural Computation, Riverside, CA, June 2, 2012. , Jun-2012
Abstracts for Journals and Proceedings	Schneider C, Case M, Soltesz I. "Enlargement and improvement of a computational model of the rat dentate gyrus." Society for Neuroscience 2012, New Orleans, LA, October 13-17, 2012. Society for Neuroscience 2012, New Orleans, LA, October 13-17, 2012. Program#/Poster#: 711.14/FFF44. Abstract available at: http://www.abstractsonline.com/Plan/ViewAbstract.aspx?mID=2964&sKey=e65a8594-02d0-4e7d-8776-5e65e7867dad&cKey=7618f57d-7015-46ca-89ee-82f2b251d9d9&mKey=70007181-01e9-4dc9-a0a2-eebfa14cd9f1 ; accessed 11/15/12. , Oct-2012

Abstracts for Journals and Proceedings	Sokolova I, Obenaus A, Nelson G, Vikolinsky R. "Radiation-Induced Suppression of LTP Is Associated with Altered Intrinsic Membrane Properties in Hippocampal CA1 Neurons." 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. Poster 8106. 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. , Jul-2012
Abstracts for Journals and Proceedings	Sokolova I, Nelson GA, Vlkolinsky R. "Basic characteristics of CA1 pyramidal neurons reflect the size of an information encoding unit." Society for Neuroscience 2012, New Orleans, LA, October 13-17, 2012. Society for Neuroscience 2012, New Orleans, LA, October 13-17, 2012. Program#/Poster#: 298.01/DDD4. Abstract available at: http://www.abstractsonline.com/Plan/ViewAbstract.aspx?sKey=8743b7a1-9dd5-4b65-9945-b969b6a07746&cKey=68719ecc-1671-4ce4-a2ce-c9cafb4099db&mKey=70007181-01c9-4dc9-a0a2-eebfa14cd9f1 ; accessed 11/15/12. , Oct-2012
Abstracts for Journals and Proceedings	Villasana LE, Dayger C, Raber J. "Effects of cranial irradiation on the cognitive function of mice: Role of reactive oxygen species." 41st Annual Society for Neuroscience Meeting, Washington D.C., November 12-16, 2011. 41st Annual Society for Neuroscience Meeting, Washington D.C., November 12-16, 2011. Abstract available at: http://www.abstractsonline.com/Plan/ViewAbstract.aspx?sKey=e86a7747-00b5-4590-9719-81b47acf460a&cKey=c06e93d4-2a0c-4e00-99bf-4d17650ba8fe&mKey={8334BE29-8911-4991-8C31-32B32DD5E6C8} ; accessed 11/15/12. , Nov-2011
Abstracts for Journals and Proceedings	Vlkolinsky R, Campbell-Beachler M, Sokolova I, Obenaus A, Nelson GA. "Functional Effects of Proton, Silicon and Iron Radiation on Synaptic Excitability in the Mouse Hippocampus." 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. Poster 8114. 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. , Jul-2012
Articles in Peer-reviewed Journals	Rosi S, Ferguson R, Fishman K, Allen A, Raber J, Fike JR. "The polyamine inhibitor alpha-difluoromethylornithine modulates hippocampus-dependent function after single and combined injuries." PLoS One. 2012;7(1):e31094. Epub 2012 Jan 27. PubMed PMID: 22299052 , Oct-2012
Articles in Peer-reviewed Journals	Villasana LE, Rosenthal RA, Doctrow SR, Pfankuch T, Zuloaga DG, Garfinkel AM, Raber J. "Effects of alpha-lipoic acid on associative and spatial memory of sham-irradiated and (56)Fe-irradiated C57BL/6J male mice." Pharmacology, Biochemistry, and Behavior. 2013 Jan;103(3):487-93. Epub 2012 Oct 7. http://dx.doi.org/10.1016/j.pbb.2012.09.021 ; PubMed PMID: 23051895 , Jan-2013
Articles in Peer-reviewed Journals	Yeiser LA, Villasana LE, Raber J. "ApoE isoform modulates effects of cranial (56)Fe irradiation on spatial learning and memory in the water maze." Behavioural Brain Research. 2013 Jan 15;237:207-14. Epub 2012 Sep 24. http://dx.doi.org/10.1016/j.bbr.2012.09.029 ; PubMed PMID: 23018124 , Jan-2013
Articles in Peer-reviewed Journals	Schneider CJ, Bezaire M, Soltesz I. "Toward a full-scale computational model of the rat dentate gyrus." Frontiers in Neural Circuits. 2012;6:83. Received: 01 Sep 2012; Paper pending published: 24 Sep 2012. Accepted: 27 Oct 2012. http://www.frontiersin.org/Neural_Circuits/10.3389/fncir.2012.00083/abstract , Oct-2012
Articles in Peer-reviewed Journals	Haley GE, Villasana L, Dayger C, Davis MJ, Raber J. "Apolipoprotein E genotype-dependent paradoxical short-term effects of (56)fe irradiation on the brain." International Journal of Radiation Oncology Biology Physics. 2012 Nov 1;84(3):793-9. Epub 2012 Mar 6. http://dx.doi.org/10.1016/j.ijrobp.2011.12.049 ; PubMed PMID: 22401921 , Nov-2012
Books/Book Chapters	Obenaus A, Mickley A, Bogo V, West BR, Raber J. "Behavioral and neurophysiological changes of radiation exposure." in "Revised Chapter 7 in the Textbooks of Military Medicine series, Medical Consequences of Nuclear Warfare." United States Army Office of the Surgeon General Borden Institute, in press, as of November 2012., Nov-2012